Collecting data in understudied language varieties: A methodological note

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Abstract

Eliciting linguistic data from speakers of non-standardized varieties presents well-known challenges. Traditional acceptability judgment tasks have been widely used to probe grammatical knowledge, but task design, such as mode of presentation, response options, and labeling, can strongly influence outcomes (Marty et al., 2020). In this study, we investigate a small group of heritage Catalan speakers in Germany and the Netherlands using a repetition-based task rather than a standard judgment task. Our findings indicate that responses to repetition tasks differ significantly from commonly known generalizations on the phenomenon at issue investigated through grammaticality judgment tasks. We conclude that repetition tasks can uncover patterns of grammatical representation that may differ from those captured by judgment tasks, and that they provide deeper insights into grammatical representation than the latter. This suggests that carefully designed elicitation tasks can mitigate common biases and more accurately reveal the grammatical representations of heritage speakers. These findings highlight both the challenges and the potential of alternative methodologies for studying underrepresented languages.

1. Introduction

Collecting data for non-standardized varieties is a challenge for linguists. The data elicitation in form of acceptability judgments has largely proved to be effective for English monolingual speakers, most notably by Sprouse, Schütze, and Almeida (2013), and Sprouse and Almeida (2017). That this might not be the whole story has, however, been noted by Marty et al. (2020), who showed that the task configuration can influence acceptability judgment tasks. This, in turn, means that the replication of the judgments can be affected by the way in which the tasks are designed. Specifically, Marty et al. identify three factors that can influence acceptability judgments: the mode of presentation, the number of response options, and the use of labels. The mode of presentation refers to whether stimuli are presented visually, aurally, or in written form, which can affect comprehension and attention. The number of response options, for example, binary "acceptable/unacceptable" choices versus Likert scales, can influence how speakers categorize linguistic phenomena, potentially masking subtle gradations in acceptability. Finally, the use of labels concerns whether linguistic terms (e.g., "grammatical," "acceptable") are provided to guide participants' responses; this can alter participants' interpretations of the task and introduce response biases. Taken together, these factors highlight that acceptability judgments are sensitive not only to the linguistic knowledge of participants but also to the design of the task itself.

In this paper, we show further evidence that different tasks might result in slightly different results. Although languages with very few speakers can be problematic for establishing generalizations (see also Leivada, D'Alessandro & Grohmann 2019, and D'Alessandro, Natvig & Putnam 2021 on this issue), we will also show that such generalizations can indeed be drawn if appropriate procedural care is applied.
Our empirical base is a small group of heritage Catalan speakers in Germany and the Netherlands. We employ a repetition-based task that indirectly taps into speakers' grammatical knowledge, showing that this approach can produce different results from acceptability judgment tasks and reveal aspects of grammatical representation that might not emerge in explicit judgments. We show that participants' responses are more closely aligned with their underlying grammatical knowledge when the task is formulated to directly engage with these representations.

2. The *yes* bias

Bilingual respondents routinely display a "yes-bias" (Polinsky, 2006) in judgment tasks, a tendency to accept sentences as grammatical or "OK", even when their internal competence would suggest otherwise. The yes-bias, i.e. the "reluctance to reject ungrammatical material" (Polinsky, 2018: 95) was first described for L2 speakers by Ellis (1991) and observed to be at work also for heritage speakers (HS) by Polinsky (2006) in an article on heritage Russian speakers in America. Romano and Guijarro Fuentes (2023) further note that this effect is magnified in metalinguistic judgment tasks for heritage grammars, where practical communication norms outweigh formal prescriptive standards. The yes-bias reflects multiple factors:

(i) Deference to the perceived authority of the researcher or stimulus provider.

(ii) Socialized tolerance of variation in bilingual communities.

(iii) Uncertainty about minority-language norms, especially when the heritage language (HL) is primarily spoken at home or in informal settings.

For HS, explicit ratings may thus underrepresent the boundaries of their active grammatical knowledge, and contribute to the "incomplete acquisition" (Polinsky, 2006, 2008; Montrul, 2002, 2008 ff, but see Putnam & Sánchez, 2013) narrative that fails to capture their dynamic competence (cf. Leivada et al., 2023).

Despite variable proficiency, HS have been shown to retain systematic representations of morphosyntax (Scontras, Fuchs & Polinsky, 2018). However, explicit judgments frequently conflict with production or comprehension data; heritage speakers may "accept" ungrammatical forms out of politeness or uncertainty, yet avoid producing such forms in spontaneous speech. Studying implicit representations requires methods less reliant on conscious reflection or explicit evaluation. This is especially important for phenomena like agreement, where intuitions can be subtle and context-sensitive. To illustrate this point, we present a small experiment on gender in HS.

2.1. Gender and number agreement in heritage speakers

Gender and number attribution and agreement are widely studied in HL works (see Polinsky 2008; Albirini et al., 2013; Rodríguez & Reglero, 2015; Lohndal & Westergaard, 2016, 2021; Jegereski & Fernandez Cuenca, 2025; Busterud et al. 2025, and others). Our study had the aim to check whether the generalizations that were found through different kinds of tests, mainly GJT, could be replicated by our population. Before going into the details of our study, a disclaimer is in order.

The term agreement refers to at least three different phenomena in linguistics: the first one is the attribution of lexical gender or number to an item; it is usually assumed (much depending on one's theory of the lexicon) that this kind of gender is assigned directly to the lexical item; for instance, the word for chair in Spanish is *silla*. This lexical item is feminine singular, and this gender is independent of any agreement relation that this word might entertain with other elements in the clause. Especially in languages like Spanish, which are heavily morphologically gender-marked, gender and number are also visible on all the items occurring within an NP; for instance, 'the old and beautiful chair' in Spanish is *la silla vieja y bonita*, where the feminine singular morpheme -a appears on all items within the NP. This phenomenon, which is similar to "spreading" the ending through the phrase, is usually referred to as Concord. The third agreement phenomenon has to do with sentential agreement: it is the sort of agreement that occurs in predicative sentences, for instance, or in secondary predication. The Spanish equivalent of English "Mary considers the chair too old" would be *María considera la silla demasiado vieja,* where *vieja* will be agreeing with la silla in the same way as it would in a simple predicative structure: *la silla está vieja.* Spanish does not have participial agreement, and hence, sentential agreement is restricted compared to, for example, Italian.

In this paper, we will be considering sentential agreement with the aim of ascertaining whether heritage speakers have a full-fledged grammar for long-distance syntactic sentential agreement, or their grammar is defective.

2.1.1. Long-distance agreement and agreement attraction

Agreement and agreement attraction, i.e. the phenomenon whereby speakers accept agreement with the linearly closest intervening element, have been widely studied, both for monolingual speakers and for HS. For our purposes, one particular study is of great importance, i.e. that by Scontras, Fuchs, and Polinsky (2018), who investigate gender and number agreement in heritage Spanish speakers in the US. Their intent was different from ours: they were trying to check whether features "probe" separately, and whether principles of featural economy are at work, reducing agreement and bundling together features that normally agree separately. What matters for us is that they found that heritage Spanish speakers present strong deviations with respect to monolinguals when it comes to acceptability judgments. SFP performed an auditory rating task: speakers were asked to rate the acceptability of a grammar on a 1-5 Likert scale.

The results are very interesting: grammaticality effects are found only in the feminine, meaning that HSs accept agreement of a feminine with a (default) masculine, but not vice versa. More than the results themselves, what is particularly interesting is that these effects are exactly the same as those found in monolingual speakers. In other words, no difference has been observed in the acceptability of long-distance agreement sentences for gender between monolinguals and HSs: what is different is that monolinguals consider the ungrammaticality of each feature (number or gender) separately, and therefore are also sensitive to the single featural mismatch, while HSs cluster the two dimensions together.

For what concerns us, the interesting results are that HSs accept ungrammatical agreement attraction by a linearly closer intervener, though it is also worthwhile considering that monolingual speakers also have the same effect (with respect to gender, not clustering). SFP reflect on the possibility that this acceptance might not stem not only from a yes-bias but also from the heavy computational load that this LDA imposes on speakers.

After considering these results, we set ourselves to check whether they would be the same if a different test were used. Like SFP, we provided auditory stimuli. In order to check whether the speakers' judgments were informed by the yes-bias, by syntactic complexity, or by the task, we decided to perform a production task in the form of repetition and correction: our informants were instructed to correct the sentences that they found ungrammatical, and repeat them correctly. If the problem were only in the length of the sentences, practically equivalent to those in SFP, we would expect the same results. If different tests yield different results, this would imply that we need to be cautious about the generalizations derived from any elicitation.

Before moving to the description of the test, one disclaimer is in order: our study was a small methodological exercise conducted under severe time constraints. This means that our findings should be regarded as a starting point for further investigation rather than as a solid generalization.

3. The test

3.1. Research questions and hypotheses

The study examined whether elicited imitation tasks could tap into heritage speakers' implicit knowledge of gender and number agreement in Catalan, and whether this method would reveal error patterns different from those observed in judgment tasks. The main predictions were:

1. Heritage speakers would "repair" ungrammatical agreement more often in repetition than acknowledge violations in judgment tasks.
2. The error profiles for number versus gender agreement attraction would help clarify which feature is more vulnerable in heritage grammars.

3.2. Methodology

3.2.1. Participants

A total of 37 informants participated in the present study. They belong to three main groups: Catalan HSs, first-generation Catalan speakers and mainland Catalan speakers. The two immigrant groups (HSs and first-generation speakers) are further subdivided by country of residence: Germany and the Netherlands, leading to four subgroups. Mainland speakers are also subdivided by generation: a younger generation, similar in age to HSs, and an older generation, similar in age to first-generation speakers. This results in six speaker subgroups in total. In this section, we will follow this latter division to better characterize the participants, but for the purpose of the analysis, we will maintain the division into three main groups.

HS, as mentioned, are divided into two groups: the group from Germany, which consists of 6 speakers (2 male; 3 female; 1 non-binary/other), aged 21-38 (mean age = 28.3; standard deviation [SD] = 6.5), and the group from the Netherlands, which consists of 3 speakers (1 male; 2 female), aged 28-43 (mean age = 36.3; SD = 7.6). The linguistic profile of the speakers is very similar. All HSs were exposed to both Catalan and German or Dutch from birth, as they all have at least one Catalan parent. The only exception is a participant from Germany who has two Catalan parents and was born in the United States. This participant moved to Germany at the age of 2, which is the age of first exposure to German. Moreover, the Catalan parents of all HSs were from the same Catalan region, which is the province of Barcelona. They all attended regular German or Dutch school, and, in parallel, three of them took Catalan lessons for a year. All participants stated being more fluent in German or Dutch than in Catalan.

Language exposure and use were evaluated according to two parameters. The first one consisted of speakers reporting the five people they most frequently talked to and the languages they used. HSs from Germany reported using German in 65.6% of their daily conversations, Catalan in 18.8%, and the remaining 15.6% with English, Spanish, and/or Portuguese[1]. HSs from the Netherlands used Dutch with 75% of their daily interlocutors and Catalan with the remaining 25%. The second parameter was asking participants to rank the extent to which they were exposed to German or Dutch and Catalan from 0 'never' to 5 'always' in different contexts (family; friends; reading; TV and series; radio, music and podcasts). Both HSs' groups reported moderate exposure to Catalan in family settings (mean = 3), but low in the other settings (Germany mean = < 2; Netherlands mean = < 1). In contrast, exposure to German or Dutch was between moderate and high across contexts, with the highest mean in friends' settings (Germany mean = 4.17; Netherlands mean = 4.7). From both parameters, it can be concluded that all participants are dominant in German or Dutch, in contrast to Catalan, as their levels of exposure and use are clearly higher in the former languages. Participants were also asked to rate their proficiency in their dominant and HLs in speaking, listening and reading on a scale from 1 'low' to 3 'high'. A summary of the results, together with those from first-generation immigrants, is shown in Table 5. As can be observed, both groups reported having an overall higher level in German or Dutch than in Catalan.

[1] This indicates that participants are multilingual rather than bilingual. While the other languages could potentially affect Catalan attainment and change, investigating their effects falls outside the scope of this study. These languages will only be considered if any results are markedly deviant.

Table 1.

Mean of self-reported ratings of Catalan, German and Dutch proficiency by speaker group

	HSs Germany (n = 6)		HSs Netherlands (n = 3)		First-generation Germany (n = 10)		First-generation Netherlands (n = 10)	
	German	Catalan	Dutch	Catalan	German	Catalan	Dutch	Catalan
Speaking	3.0	2.2	3.0	2.0	2.7	3.0	2.0	2.9
Understanding	3.0	2.8	3.0	2.3	2.8	3.0	2.4	3.0
Reading	2.8	2.0	3.0	1.7	2.7	3.0	2.2	3.0
Mean	2.9	2.3	3.0	2.0	2.7	3.0	2.2	3.0

The first-generation immigrants comprise 20 Catalan speakers: 10 living in Germany (10 female), aged 28-63 (mean age = 46.5; SD = 9.7), and 10 living in the Netherlands (2 male; 8 female), aged 33-63 (mean age = 49.0; SD = 9.9). They all left Catalonia and, in one case, the Balearic Islands between the ages of 18 and 34, and have lived for at least 10 years in Germany (mean length of residence = 19.5; SD = 7.3) or in the Netherlands (mean length of residence = 17.2; SD = 7.0). Most of them are from the same Catalan province as the parents from the HSs, which is Barcelona, with two being from Girona and one from the Balearic Islands. They were all schooled in Catalan, except for the two oldest participants. This is because during the Spanish Francoist dictatorship (1939-1975) and the years following its fall, education in Catalan was prohibited. Of the twenty speakers, eighteen reported having Catalan as their most fluent language, one reported German, and one reported English, followed by Dutch.

The same parameters used to evaluate language exposure and use in HSs were applied to this group. Regarding the first parameter, involving the people they most frequently interact with and the languages they use with them, first-generation speakers from Germany reported using German in 52.5% of their daily conversations, Catalan in 27.9%, and Spanish and/or English in the remaining 19.6%. Instead, first-generation speakers from the Netherlands used Dutch in 24.1% of their daily interlocutors, Catalan in 44.4% and Spanish and English in the remaining 31.5%. As for the second parameter, which measures language exposure across different contexts, both groups had Catalan exposure in the family domain as the highest (mean = 3.7). However, while participants from Germany reported a higher exposure to German than to Catalan in all the other settings, participants from the Netherlands had a higher exposure to Catalan than to Dutch in all contexts except with friends (mean = 1.8). This contrast between language exposure and use between first-generation speakers living in Germany and in the Netherlands is reflected in their self-rating proficiency in the dominant language: it is higher in German than in Dutch, as illustrated in Table 5.

The mainland Catalan speakers' groups constitute the control groups. They broadly matched first-generation immigrants and HSs in age, which means they also constitute two different generations. The younger speakers were aged 18-25 (n = 4; 2 male, 1 female, 1 non-

binary/other), and the older speakers were aged 51-60 (n = 4; 2 male, 2 female). Moreover, they are also from the same Catalan region as first-generation speakers and HSs' Catalan parents: the province of Barcelona. They all have Catalan as their dominant language, with only one reporting being equally dominant in Catalan and Spanish.
Recruitment of first-generation and HSs was carried out via social media and through snowball sampling. Mainland speakers were primarily recruited via the personal network of the experimenter.

3.2.2. Stimuli

The task consisted of 30 stimuli: 18 experimental items, 4 control sentences, and 8 fillers. Because the task was part of a larger study (Colina Fortuny, 2025), we used the filler items to test the phenomenon of interest, namely, sentences that can trigger agreement attraction effects. The stimuli were adapted from the materials of Fuchs et al. (2015), later used in Scontras, Fuchs and Polinsky (2018). With the aim of using a structure where a noun intervenes between the source and target of agreement, and where all agreeing elements are marked for both gender (masculine vs feminine) and number (singular vs plural), they designed sentences with a small clause, with the following syntactic structure:

(1) (Subject) Verb [SC NP1 Prep NP2 Adv ADJ] … (Scontras, Fuchs & Polinsky, 2018)

As it can be observed in (1), each small clause includes a noun (NP1) that is modified by a prepositional phrase containing a local noun (NP2). The small-clause subject (NP1) agrees with the predicative adjective or participle (ADJ), with the NP2 in between, as a distractor. The sentence is thus only grammatical if agreement of both number and gender takes place between the NP1 and the ADJ.

The stimuli were created by manipulating the number and gender of NP1, NP2 and ADJ, which resulted in grammatical and ungrammatical sentences. Of the 8 stimuli sentences, 4 targeted gender (mis)matches and 4 targeted number (mis)matches. Regarding the gender sentences, half were designed so that the ADJ agreed in both gender and number with the NP1 (grammatical), while matching NP2 only in number. In the other half, the ADJ agreed in both gender and number with the NP2 (ungrammatical), while matching NP1 only in number. In all cases, the two nouns differed in gender but shared the same number, as illustrated in (2).

(2) a. El nen considera l' article de la
the kid consider.PRS.3SG the.M/F.SG article.M.SG of the.F.SG
revista completament tràgic.
magazine.F.SG completely tragic.M.SG
b. *El nen considera l' article de la
the kid consider.PRS.3SG the.M/F.SG article.M.SG of the.F.SG
revista completament tràgica.
magazine.F.SG completely tragic.F.SG
Intended: 'The boy considers the article in the magazine completely tragic.'

The same was done with the sentences targeting number. Two were grammatical, with the ADJ agreeing with NP1 in gender and number, and only in gender with NP2. The other two were ungrammatical, with the ADJ agreeing with the NP2 in gender and number, and only in gender with the NP1. Again, in all cases, the two nouns differed in number but shared the same gender, as in (3).

(3) a. Considero la campana de les escoles
consider.PRS.1SG the.F.SG bell.F.SG of the.F.PL school.F.PL
excessivament sorollosa.
excessively noisy.F.SG
b. *Considero la campana de les escoles
consider.PRS.1SG the.F.SG bell.F.SG of the.F.PL school.F.PL
excessivament sorolloses.
excessively noisy.F.PL
Intended: 'I consider the school bell excessively noisy.'

This design allowed us to test whether HSs made more errors in number or in gender, whether these errors were driven by agreement attraction, and whether this pattern was consistent across grammatical and ungrammatical sentences.

Both the materials and the language background questionnaire can be found in the Appendix.

3.2.3. Task

The elicited imitation task was administered via a PowerPoint presentation. In each trial, participants first heard a sentence, followed by a short pause during which numbers, figures and letters were displayed on the screen. At the end of this pause, a beep indicated that they could repeat the sentence. This design was chosen because the rationale of this task is that sentence recall does not produce a passive copy but measures implicit linguistic knowledge (Bowles, 2011; Ellis, 2005; Erlam, 2006). Introducing the pause has further been shown to reduce speakers' reliance on working memory (Kostromitina & Plonsky, 2022).

The task included two practice sentences (one grammatical, one ungrammatical) to familiarize participants with the procedure. The stimulus sentences were elicited by the experimenter, a mainland Catalan speaker, and were presented in a randomized order to prevent confounding learning effects. Sentences were repeated once, and only twice if requested.

3.2.4. Procedure

The task was administered online via Google Meet. Participants were instructed to use a computer in a quiet environment with a stable internet connection. Sessions were audio-recorded using OBS Studio, and later transcribed and coded for analysis. At the end of each session, information was collected on participants' demographic data and linguistic history via the survey platform Qualtrics (https://www.qualtrics.com). The demographic information included age, gender, place of birth, and place of residence. The linguistic data was divided into two blocks: the first covered general information about language acquisition, fluency, and use, and the second focused on Catalan and Dutch or German, with questions about schooling, self-rated proficiency and exposure.

Participants took part in the experiment on a voluntary basis and provided oral consent at the beginning of each session. The study received positive post-hoc advice from the Linguistics Chamber of the Faculty Ethics assessment Committee of the Faculty of Humanities (FEtC-H) (reference number: 25-076-02).

At the beginning of the session, participants were provided with a detailed explanation of the task procedure and were explicitly informed that some sentences might sound strange or unusual, and that they should repeat them as naturally as possible. During the practice phrase, if a participant failed to correct the ungrammatical sentence, they were asked if they had noticed anything strange and how they would normally say it. If the participant did correct it, the experimenter just confirmed that this was the intended approach. This was done because in the pilot study, participants tended to repeat ungrammatical sentences without correcting them, even when noticing the errors, arguing that the instructions had been to simply repeat the sentences. After completing the two practice trials and prior to beginning the main task, participants were given the opportunity to ask questions.

During the main task, and since the internet connection was not stable across sessions, the experimenter always made sure that the participants were clearly hearing the stimuli and instructed them to ask for repetitions in case there were words they could not hear well. Despite these measures, one participant was unable to complete the task due to auditory difficulties.

3.3. Results

This section presents the results of the elicited imitation task. We begin with an overview of speakers' overall performance, followed by a general overview of the error types, and end with an analysis of agreement attraction effects based on the features of the head noun. Given the limited number of stimuli, we will report the raw counts without statistical results.

3.3.1. Overall performance

The performance of the three participant groups in repeating the grammatical and ungrammatical stimuli is summarized in Table 2. For grammatical sentences, responses were considered correct if the resulting sentence was grammatical, whether the participants produced an exact repetition or modified the number and/or gender features of NP1, NP2 and/or ADJ while remaining grammatical. For ungrammatical sentences, responses were considered correct only when participants modified the sentence to make it grammatical. Instead, exact repetitions or changes that lead to an ungrammatical sentence were counted as incorrect. Note that only those sentences that maintained the structure in (1) were kept in the analysis. Changes that lead to a different syntactic construction were discarded.

Table 2.

Performance in repeating grammatical sentences and correcting ungrammatical sentences

	Mainland speakers (n= 8)			First-generation speakers (n = 19)			HSs (n = 9)		
	Count	M (%)	SD	Count	M (%)	SD	Count	M (%)	SD
Repeat grammatical	32/32	100.0	0.0	73/75	96.9	9.3	26/33	72.2	42.3
Correct ungrammatical	30/32	93.8	11.6	69/76	90.8	17.1	24/29	82.8	22.7

Overall, Table 2 shows that mainland speakers, independently of sentence type, achieved the highest accuracy. First-generation speakers were less accurate in their responses, but with comparable results to mainland speakers. In contrast, HSs was the group with a lower accuracy, especially when repeating grammatical sentences, where their performance varied considerably.

To better understand participants' performance, Table 3 presents the nature of their responses for each sentence type. Specifically, it shows whether they produced exact repetitions or changed them, and whether those changes resulted in grammatical or ungrammatical outputs. Note that while exactly repeating a grammatical sentence yields to a grammatical output, exactly repeating an ungrammatical sentence does not.

Table 3.

Response types for grammatical and ungrammatical sentences

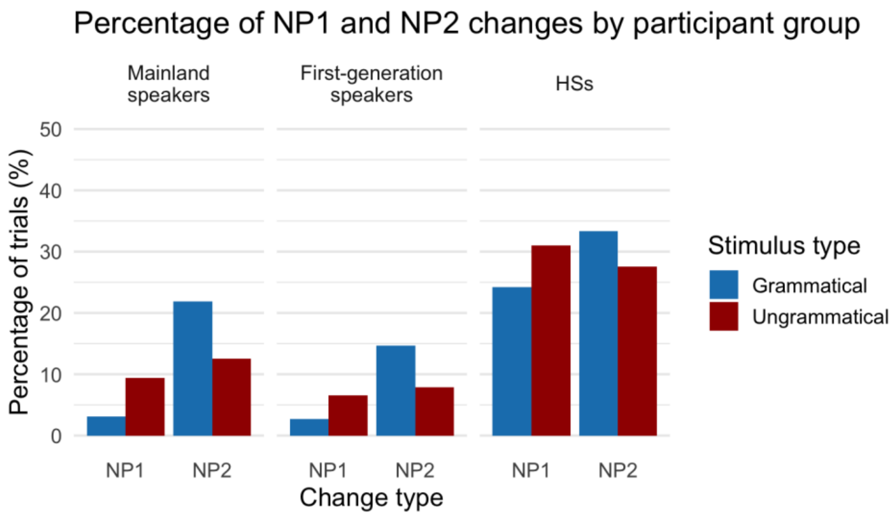
	Mainland speakers			First-generation speakers			HSs		
	Count	M (%)	SD	Count	M (%)	SD	Count	M (%)	SD
Grammatical sentence									
Exact repetition	24/32	75.0	18.9	58/75	76.3	27.0	15/33	41.7	41.5
Changed grammatical	8/32	25.0	18.9	15/75	20.6	21.4	11/33	30.6	34.9
Changed ungrammatical	0/32	0.0	0.0	2/75	3.1	9.3	7/33	27.8	42.3
Ungrammatical sentence									
Exact repetition	2/32	6.3	11.6	6/76	7.9	14.6	2/29	9.4	18.6
Changed grammatical	30/32	93.6	11.6	69/76	90.8	17.1	24/29	80.2	22.7
Changed ungrammatical	0/32	0.0	0.0	1/76	1.3	5.7	3/29	10.4	14.6

Table 3 restates what is observed in Table 2, but in greater detail. Mainland speakers show near-ceiling performance: they produced a high percentage of exact repetitions of grammatical sentences and corrected almost all ungrammatical sentences. First-generation also performed well: the percentage of exact repetitions of grammatical sentences was high, and they correctly changed most of the ungrammatical sentences, with their most frequent error (although at a very low percentage) being the exact repetition of ungrammatical sentences. Finally, HS displayed greater variability: for grammatical sentences, they most often produced grammatical outputs, but they were less accurate in producing exact repetitions. For ungrammatical sentences, they generally changed them into grammatical ones, and showed similar rates of errors in exactly repeating them or changing them for an ungrammatical sentence.

Finally, an important aspect to consider for the analysis is whether participants only changed the number and gender features of the adjective, as expected, or whether they also modified the features of NP1 and/or NP2 in their production. Figure 1 illustrates these changes. HSs produced the highest proportion of changes overall, followed by mainland speakers, with first-generation speakers producing the lowest proportion. Interestingly, across groups, NP2 was more often changed in grammatical than ungrammatical stimuli, while NP1 showed the opposite pattern, being more often changed in ungrammatical than grammatical stimuli.

Figure 1.

NP1 and NP2 changes grouped by stimulus type.



3.3.2. Agreement errors type

In this section, we examine the agreement errors in gender and number. These comprise agreement attraction errors, which involve these phi-features, as well as gender errors that follow a masculine-default strategy or a shape-based strategy. The masculine default strategy consists of assigning masculine gender to all nouns, regardless of whether they are masculine or feminine. The shape-based strategy consists of assigning gender on the basis of the noun's phonological, morphological or orthographic form of the noun (e.g., in Spanish, nouns ending in -o are associated with masculine and those ending in -a to feminine). The distribution of gender and number error types is summarized in Table 4.

Table 4.
Gender and number error types (counts and proportions).

	Mainland speakers		First-generation speakers		HSs	
	Count	%	Count	%	Count	%
Gender errors						
Agreement attraction	1/64	1.6	5/151	3.3	3/62	4.8
Masculine default	0/64	0.0	1/151	0.7	2/62	3.3
Shape-based	0/64	0.0	0/151	0.0	4/62	6.5
Number errors						
Agreement attraction	1/64	1.6	3/151	2.0	1/62	1.6

Overall, gender and number errors were limited across speaker groups, with HSs producing more errors, although at low rates. Across groups, gender-related errors were more frequent than number ones. Among the error subtypes, agreement attraction was the most common: it was the only error type produced by mainland speakers and the most frequent one among first-generation speakers. For HSs, however, shape-based errors showed the highest percentage.

3.3.3. Agreement attraction

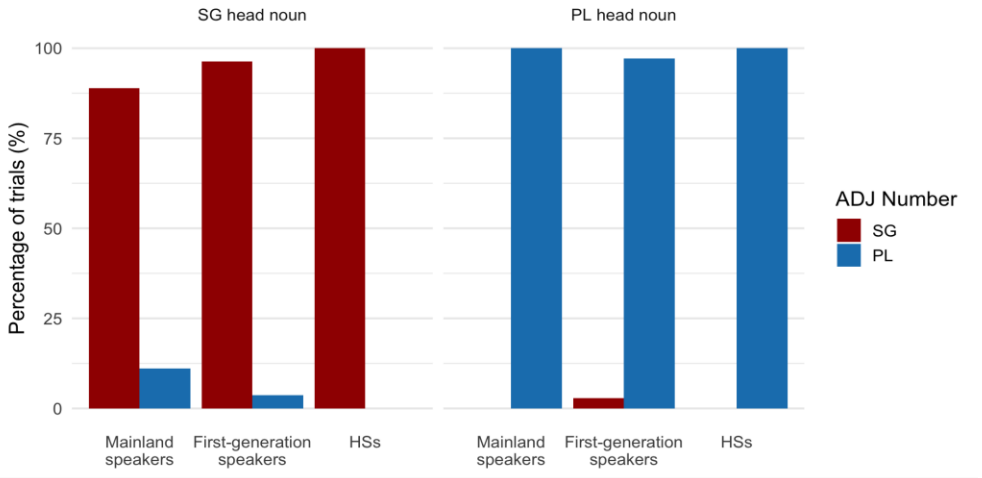
This section follows Fuchs et al. (2015) and Scontras, Fuchs and Polinsky (2018) in examining agreement attraction effects for number and gender, with the analysis organized by the features of the head noun (NP1), which determines agreement. In order to compare the results of these studies to those of the present one, we will only use those sentences where participants only changed the features of the ADJ in their repetitions, and not those of the NP1 and/or NP2. Based on this criterion, Figure 2 shows participants' production of adjective number as a function of NP1 number, while Figure 3 presents their production of adjective gender as a function of NP1 gender.

Singular head noun To avoid possible effects of gender mismatches, gender was kept constant across NP1, NP2, and ADJ. To elicit agreement attraction, NP2 was assigned plural number, in contrast to the singular NP1. Thus, all cases in which the ADJ appeared in the plural reflect agreement attraction. As shown in Figure 1, all groups produced a high percentage of singular adjectives, consistent with the head noun. Performance was at ceiling in the case of HSs. Both mainland and first-generation speakers produced plural adjectives due to agreement attraction effects, but the percentage was low.

Plural head noun As in the singular head noun condition, gender was fixed across NP1, NP2, and ADJ. In this case, the NP2 was singular, in opposition to the head noun. All groups produced a very high percentage of plural adjectives, in agreement with the NP1. Performance was at ceiling for mainland and HSs, with only a few number errors produced by first-generation speakers.

Figure 2.

Speakers' production of singular and plural adjectives grouped by NP1 number.

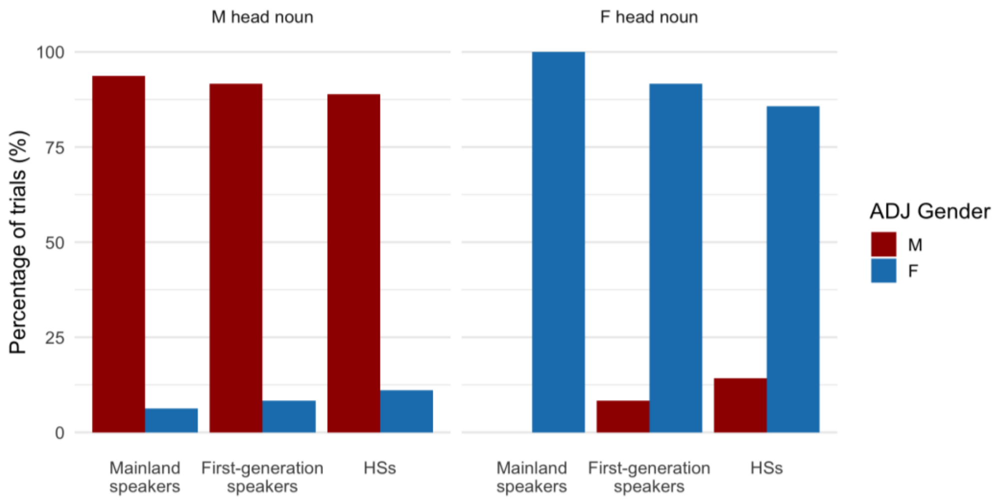


Masculine head noun To avoid possible effects of number mismatches, number was kept constant across NP1, NP2 and ADJ. To trigger agreement attraction, NP2 was assigned feminine gender, in contrast to the masculine NP1. Consequently, all the cases where the ADJ appeared in the feminine reflect agreement attraction. As it can be observed in Figure 2, all three groups produced a high percentage of masculine adjectives, with only a small proportion of feminine errors.

Feminine head noun As with the analysis of masculine head nouns, number was fixed across NP1, NP2 and ADJ. In this case, NP2 was masculine, in contrast with the feminine head noun. Across speaker groups, the percentage of correct agreement between the NP1 and the ADJ was high, reaching ceiling levels for mainland speakers. Both HSs and, to a lesser extent, first-generation speakers produced agreement attraction errors by using a masculine ADJ, although the percentages are low.

Figure 3.

Speakers' production of masculine and feminine adjectives grouped by NP1 gender.



4. Discussion and conclusions

This study departs from the widely recognized challenge that the design and selection of experimental tasks crucially shape the data elicited from speakers of non-standardized varieties. Acceptability judgment tasks, although widely used to probe grammatical knowledge, have been shown to be sensitive to task parameters in ways that can significantly affect outcomes (Marty et al., 2020). To address this issue, we compare two methodologies—acceptability judgments and elicited imitation—to explore heritage speakers' grammatical representations.

Our starting point is the study by Scontras, Fuchs, and Polinsky (2018), who investigated gender and number agreement in Spanish heritage speakers in the U.S. using an acceptability judgment task. Their results suggested differences between heritage speakers and monolinguals: while monolinguals rated sentences with a single feature mismatch (gender or number) as more acceptable than those with mismatches in both features, heritage speakers rated both types of violations equally. This was interpreted as evidence that, in heritage grammars, gender and number features are bundled and valued together. Moreover, they found asymmetries in feature content: both monolinguals and heritage speakers tolerated default masculine adjectives with feminine nouns, but only heritage speakers accepted singular adjectives with plural nouns. This was taken as evidence of feature loss in number, with only plural specified in heritage grammars.

To test whether such findings may be task-dependent, we conducted an elicited imitation task, which is argued to tap into implicit grammatical knowledge. Following the stimulus structure of Scontras et al. while focusing specifically on feature content, we manipulated gender and number separately across the head noun (NP1), intervener (NP2), and adjective (ADJ), controlling for potential confounds. Contrary to previous findings, our results revealed no evidence of feature loss: heritage Catalan speakers made very few errors, at rates comparable to first-generation and mainland speakers, and they frequently corrected ungrammatical sentences regardless of the features involved. These findings indicate that heritage grammars retain values for both number and gender.

Taken together, these outcomes highlight two key points. First, they challenge deficit-based accounts by showing that heritage speakers' feature representations can be robust when tested with tasks targeting implicit knowledge. Second, they demonstrate that methodological choices strongly affect the patterns we observe: while judgment tasks may underestimate heritage competence, repetition-based tasks can reveal richer aspects of grammatical representation. We therefore argue for a multi-method approach in heritage language research, where complementary methodologies are combined to provide a more comprehensive picture of heritage grammars. Only through such an approach can robust theoretical generalizations be drawn and simplistic deficit models be overcome.

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Appendix

I. Background questionnaire

I.1. *Questionnaire for first-generation speakers and HSs*

General information

Q1. Age span

Q2. Gender

- Male
- Female
- Other
- Prefer not to say

Q3. Place of birth

- Catalonia, the Valencian Community, the Balearic Islands, or Andorra
- Germany/Netherlands

If Option 1: Q3.1. In which city did you grow up?

If Option 2: Q3.2. Where are your parents from?

Q4. Please indicate the countries or regions where you have lived and the number of years you have lived in each.
(Example: Catalonia – 30 years, Netherlands – 10 years)

	1	**2**	**3**	**4**	**5**
Country or region					
Years					

Q5. Have you lived in these countries continuously or in separate periods (for example, 10 consecutive years or 5 + 5 years separately)?

- Continuously
- In separate periods

If Option 2: Q5.1. Explain your answer.

Linguistic background

Q1. Please list all the languages you know in order of fluency, that is, ranked according to how comfortable you feel speaking them. Place first the one you speak with the greatest ease and last the one you find the most difficult to speak.

1	**2**	**3**	**4**	**5**

Q2. Please list all the languages you know in the order in which you acquired them, that is, in the order you learned them. Also, indicate the age (approximately) at which you learned each language.

(If you learned two languages at the same time, list them in separate boxes and indicate the same age for both.)

	1	**2**	**3**	**4**	**5**
Language					
Age at which you learned it					

Q3. Please list the five people you speak with most often. Indicate the language you usually use with each of them.
(*Note:* You do not need to include names, only the type of relationship. Example: son – Catalan, friend 1 – English.)

	1	**2**	**3**	**4**	**5**
Person					
Language					

Catalan language background

Q1. Have you received education in Catalan (as the language of instruction at school, high school, or university) or studied Catalan as a subject/foreign language?

- Yes
- No

If Yes: Q1.1. What type of education in Catalan have you received?

- Language school
- Education in Catalan (school, high school, university, etc.)
- Self-study
- Other: ____

Q1.2. For approximately how many years have you studied in Catalan or studied the Catalan language?

Q2. On a scale from 1 to 3, select your level of proficiency in speaking, understanding, and reading Catalan.
(*Note:* 1 means 'low', 2 'intermediate', 3 'high')

Speaking:		**Understanding spoken language:**		**Reading:**	

Q3. Please rate to what extent you are currently exposed to Catalan in the following contexts.

(*Note:* 0 means ‘never’ and 5 ‘always’)

Interacting with friends		**Watching TV, YouTube, TikTok videos, Instagram, etc.**	
Interacting with family			
Reading		**Listening to the radio, podcast, music**	

Dutch/German language background

Q1. Have you received education in Dutch/German (as the language of instruction at school, high school, or university) or studied Dutch/German as a subject/foreign language?

- Yes
- No

If Yes: Q1.1. What type of education in Dutch/German have you received?

- Language school
- Education in Dutch/German (school, high school, university, etc.)
- Self-study
- Other: ____

Q1.2. For approximately how many years have you studied in Dutch/German or studied the Dutch/German language?

Q2. On a scale from 1 to 3, select your level of proficiency in speaking, understanding, and reading Dutch/German.
(*Note:* 1 means ‘low’, 2 ‘intermediate’, 3 ‘high’)

Speaking:		**Understanding spoken language:**		**Reading:**	

Q3. Please rate to what extent you are currently exposed to Dutch/German in the following contexts.
(*Note:* 0 means ‘never’ and 5 ‘always’)

Interacting with friends		**Watching TV, YouTube, TikTok videos, Instagram, etc.**	
Interacting with family			
Reading		**Listening to the radio, podcast, music**	

I.2. *Questionnaire for mainland speakers*

Q1. Age span

Q2. Gender

- Male
- Female
- Other
- Prefer not to say

Q3. Which city or town are you from (within Catalonia, the Valencian Community, the Balearic Islands, or Andorra)?

Q4. Indicate the highest level of education you have completed in Catalan

- Primary education
- Lower secondary education
- Upper secondary education
- Vocational training
- Undergraduate studies
- Postgraduate studies
- Master’s degree
- PhD
- Other: ____

Q5. Is Catalan your dominant language?
(*Note:* ‘Dominant’ refers to the language you use most frequently in your daily life.)

- Yes
- No
- Other: ____

II. Stimuli

Item	NP1		NP2		ADJ		Test tokens
	Gender	Number	Gender	Number	Gender	Number	
1a	M	SG	F	SG	M	SG	El nen considera l’article de la revista completament tràgic. ‘The boy considers the article in the magazine completely tragic.’
1b	M	SG	F	SG	F	SG	*El nen considera l’article de la revista completament tràgica. Intended: ‘The boy considers the article in the magazine completely tragic.’
2a	F	PL	M	PL	F	PL	El nen considera les notícies dels diaris completament tràgiques. ‘The boy considers the news in the newspapers completely tragic.’
2b	F	PL	M	PL	M	PL	*El nen considera les notícies dels diaris completament tràgics. Intended: ‘The boy considers the news in the newspapers completely tragic.’
3a	F	SG	F	PL	F	SG	Considero la campana de les escoles excessivament sorollosa. ‘I consider the school bell excessively noisy.’
3b	F	SG	F	PL	F	PL	*Considero la campana de les escoles excessivament sorolloses. Intended: ‘I consider the school bell excessively noisy.’
4a	M	PL	M	SG	M	PL	Considero els rellotges de l’institut excessivament sorollosos. ‘I consider the clocks in the school excessively noisy.’

4b	M	PL	M	SG	M	SG	*Considero els rellotges de l'institut excessivament sorollós. Intended: 'I consider the clocks in the school excessively noisy.'